

WORK EXPERIENCE

CrossRefHVAC · crossrefhvac.com

2026–Present

- *Founder & Software Engineer, Barrett Technologies LLC*

Python, PostgreSQL/Supabase, Parquet, Next.js, TypeScript, pdfplumber, pytest, Vercel

HVAC/R equipment data platform, publicly launching as SpecTrace: manufacturer documentation and product APIs normalized into cross-referenced, queryable data with per-field provenance, the audit trail that keeps machine extraction verifiable at scale.

- Built a test-driven Python extraction pipeline (378 tests) turning dense manufacturer PDFs (spec tables, nomenclature systems, model families) into structured product data, with change-tracked acquisition of 100+ source documents via a lifecycle state machine.
- Mapped an undocumented manufacturer product API using agent-assisted discovery, enumerating its full parameter space (680+ configurations, three brand catalogs, 27K+ SKUs) with convergence probes verifying complete coverage.
- Built the ground-truth verification layer: pipeline outputs checked against an independent authoritative product index, with precision metrics and a persistent human-adjudication ledger; unverified data is blocked from promotion.
- Designed an OLAP fact-table schema in PostgreSQL normalizing extracted specs into queryable facts and dimensions, validated against human-curated ground truth, with versioned columnar Parquet snapshots.
- Built human-in-the-loop review tooling: a PDF-overlay UI inspecting extraction results against the source document, with each ruling persisting to the adjudication ledger so decisions are made once and reapplied automatically.
- Serving the dataset through a REST API and an MCP server with per-field provenance (every spec value traceable to its source document and page), making the data directly queryable by AI agents and HVAC-focused chat tools.

No Plastic, No Problem. · noplasticnoproblem.com

2023–Present

- *Founder*

Next.js, React, TypeScript, Tailwind, Sanity CMS, n8n, Claude API, PostHog, Sentry, Supabase, Vercel

Production affiliate content platform serving 10,000+ monthly users with 50+ brand partnerships.

- Built and operate a production affiliate platform serving 10,000+ monthly users, with a custom TypeScript/Sanity pipeline managing 200+ content entities and a bidirectional Notion integration powering the full editorial workflow.
- Built an end-to-end content automation pipeline using n8n, DataForSEO, and the Claude API, cutting SEO brief generation from hours to automated output and supporting a 3x increase in content throughput.
- Reduced CDN request volume by 85% and eliminated compute cost spikes by migrating to webhook-driven ISR revalidation, with automated search index regeneration triggered on every content publish.
- Built a custom MCP server connecting Sanity and Notion as a live knowledge graph, giving AI workflows real-time access to article inventory and brand relationships for fully grounded content generation.

- **Haven Technologies**

Fullstack Software Engineer

New York, NY

June 2022 - August 2023

- Redesigned and rebuilt an insurance policy viewer into a scalable SaaS product using Angular, PostgreSQL, Node.js, and Express, with test coverage via Mocha, Chai, and Sinon.
- Implemented NgRx for state management and centralized API call patterns, improving consistency and debuggability across the frontend codebase.

- **Allvue Systems**

Application Engineer, Operations

New York, NY

July 2020 - May 2022

- Resolved TSQL, XML, and XSLT errors in custom portfolio management software, improving reliability for B2B clients.
- Wrote scripts to fix trade data and payment processing inconsistencies, ensuring accuracy in compliance assessments.

EDUCATION

- **Stevens Institute of Technology**

Bachelor of Science in Computer Science

Hoboken, NJ

September 2016 - May 2020

TECHNICAL SKILLS

Programming

TypeScript, JavaScript, Python, SQL, HTML/CSS

Frameworks & Libs.

React, Next.js, Angular, Redux, NgRx, RxJS, Express, Node.js, Tailwind CSS

Tools

Git, GitHub, Vercel, Sanity, n8n, Sentry, PostHog, pytest, Vitest, pdfplumber, Docker, AWS, Jira

Data

PostgreSQL, Supabase, Parquet, MongoDB, Oracle, MySQL